

Question 1: What is Machine Learning?

Introduction:

Machine Learning is a subset of artificial intelligence that enables computers to learn from data without being explicitly programmed. It involves algorithms that automatically improve through experience and exposure to training data. ML builds mathematical models based on sample data to make predictions or decisions without rule-based programming.

Types:

There are three main types of machine learning. Supervised Learning trains on labeled data to map inputs to known outputs · used in classification and regression. Unsupervised Learning finds hidden patterns in unlabeled data · used in clustering and dimensionality reduction. Reinforcement Learning trains agents through reward and penalty signals to maximize cumulative reward · used in robotics and game-playing systems.

Applications:

Machine learning is widely applied in image and speech recognition, natural language processing, recommendation systems, medical diagnosis, fraud detection, and autonomous vehicles. It powers voice assistants, spam filters, financial forecasting, and drug discovery pipelines.

Conclusion:

Machine learning has fundamentally transformed computing by enabling systems to learn from experience rather than explicit programming. Its ability to extract patterns from large datasets makes it indispensable in modern technology across every industry.

Question 2: Explain Neural Networks

Introduction:

Neural networks are computational models inspired by the structure and function of the human brain. They consist of layers of interconnected artificial neurons that process information, learn feature representations, and produce outputs for complex tasks like image recognition and natural language understanding.

Architecture:

A neural network is organized into three types of layers: an input layer that receives raw data, one or more hidden layers that extract progressively abstract features, and an output layer that produces the final prediction. Each neuron computes a weighted sum of its inputs, adds a bias, and applies a non-linear activation function such as ReLU, Sigmoid, or Tanh. Deep neural networks with many hidden layers are called deep learning models.

Training:

Neural networks are trained using backpropagation combined with gradient descent optimization. During the forward pass, data flows through the network to produce a prediction. The loss function measures prediction error. Backpropagation computes the

gradient of the loss with respect to each weight using the chain rule. The optimizer (SGD, Adam) then updates weights to minimize the loss over many training iterations.

Conclusion:

Neural networks are universal function approximators capable of learning arbitrarily complex mappings from input to output. They form the backbone of modern deep learning and have achieved state-of-the-art results in computer vision, NLP, speech recognition, and generative AI applications.